# M. Priyambica

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## OBJECTIVE

To associate with an innovative organization as a Software Engineer, leveraging strong problem-solving abilities, proficiency in database management (SQL), and full-stack development skills to build scalable software solutions.

## EDUCATION

**Stanley College of Engineering & Technology for Women** **2023 – 2027**
*B.E. in Computer Science & Engineering* CGPA: 8.5

**Narayana College** **2021 – 2023**
*Class XII* Aggregate: 80%

**St Ann's High School** **2020 – 2021**
*Class X (SSC)* Aggregate: 90%

## EXPERIENCE

**Fractal Analytics** | *Project Intern* **May 2026 – Jul 2026**

- Engineered backend architectures for a Request for Proposal (RFP) tool using Python, building scalable RESTful APIs.

## PROJECTS

**Vehicle Counting at Traffic Signal** **(Completed)**

- Developed a traffic monitoring system using Python and OpenCV for real-time vehicle detection and counting.
- Engineered object tracking algorithms using region-of-interest (ROI) filtering and bounding box logic.
- Implemented background subtraction and contour detection to accurately isolate moving vehicles in video streams.
- **Technologies:** Python, OpenCV, NumPy, Computer Vision.

**Fake Media Detector** **(Completed)**

- Developed a deep learning-based detection system using **Python** and neural networks to accurately identify manipulated images and deepfake media.
- Engineered a computer vision pipeline utilizing **OpenCV** and Convolutional Neural Networks (CNNs) to analyze pixel-level artifacts and classify media authenticity.
- **Technologies:** Python, TensorFlow/PyTorch, OpenCV, Convolutional Neural Networks (CNNs), Deep Learning.

**RAG Document Assistant** **(Completed)**

- Developed a Retrieval-Augmented Generation (RAG) system using **Python** and **LangChain**, engineering a hybrid retrieval pipeline that combines **Chroma** vector search and **BM25** keyword indexing to enable precise question-answering on custom documents.
- Integrated the **Google Gemini** API with cross-encoder reranking to generate grounded, hallucination-free answers with exact source citations, served via a custom standard-library HTTP backend.
- **Technologies:** Python, LangChain, Google Gemini API, Chroma (Vector Database), BM25, Local Embeddings.

## TECHNICAL SKILLS

**Languages:** Python, C, JavaScript
**Databases:** Relational Databases (RDBMS), PostgreSQL, MySQL, SQLite
**Libraries:** Scikit-learn, Pandas, TensorFlow, NumPy, LangChain, LangGraph
**Web Technologies:** HTML5, CSS3, Streamlit
**Tools & Platforms:** Git, GitHub, VS Code, Postman

## CERTIFICATIONS

- Working with Data – **Microsoft Learn**
- Artificial Intelligence Fundamentals with Capstone Project – **IBM**
- Computer Organization and Architecture – **MindLusters**